

Answer the following two questions

- (i) A processor pipelines instructions with 4 stages: Fetch Instruction (F), Decode (D), Execute (E), and Store Result (S). Give the type of the pipeline conflict that may occur when a read operation takes place before a write operation is completed; Use a timing diagram for instruction pipelining to indicate the clock cycle(s) during which this conflict may occur.

(5 marks)

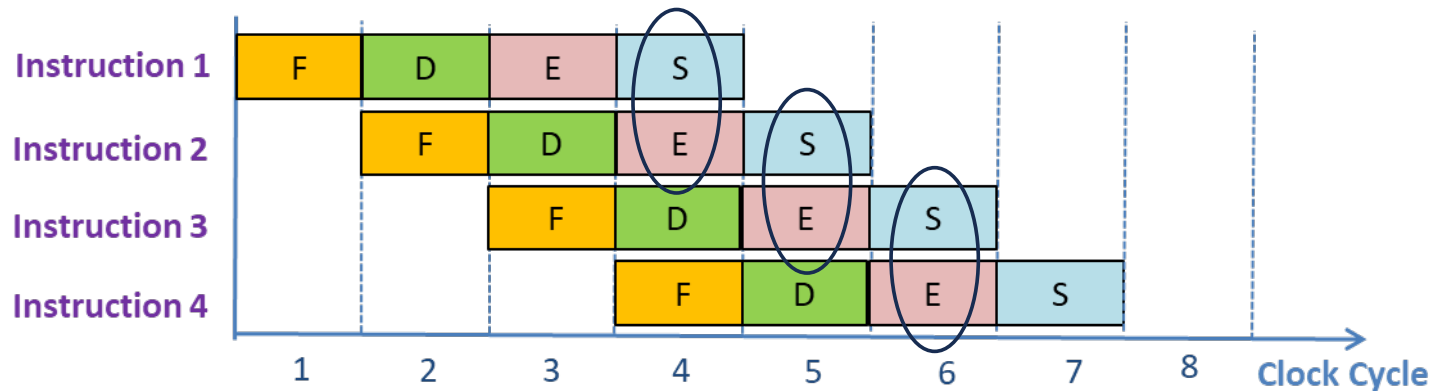
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(5 marks)

(i) Data Dependency Conflict

(2 marks)



(3 marks)

PYP related to this part

Consider a processor with 4 pipeline stages: Fetch Instruction (F), Decode (D), Execute (E) and Store (S). Assume that

- Branch target address is calculated at the execute stage
- Instruction length for every instruction is one word long
- Each pipeline stage takes 1 cycle to complete
- No resource conflicts
- Delayed branching is enabled.
- This processor is not an ARM processor. All instructions will update the relevant status flags.

With reference to the code in Figure Q1,

- (i) List all the instruction pairs and the corresponding register(s) that have data dependency conflict. E.g. if R1 in I2 and I6 has data dependency issue, then a valid answer is "I2 and I6 (R1)".

(5 marks)

	MOV	R0, #10	;I1	; R0 ← #10 (Loop counter for 10 iterations)
	MOV	R1, #0x800	;I2	
	MOV	R2, #0x100	;I3	
Loop	LDR	R4, [R2], #1	;I4	; (Step1) R4←[R2]. (Step2) R2←R2+1.
	ADD	R5, R4, R0	;I5	; R5←R4+R0
	STR	R5, [R1], #1	;I6	; (Step1) [R1] ←R5. (Step2) R1←R1+1
	BNE	Loop	;I7	; Branch on Not Equal. (Z ≠ 1)
	SUB	R0, R0, #1	;I8	; R0 = R0 - 1
	NOP		;I9	
	STR	R1, [R1]	;I10	[R1] ←R1
	STR	R2, [R2]	;I11	

Figure Q1

PYP related to this part

(i)

- I3 and I4 [R2]
- I4 and I5 [R4]
- I5 and I6 [R5]

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	MOV	R0, #10	;I1	; R0 ← #10 (Loop counter for 10 iterations)
	MOV	R1, #0x800	;I2	
	MOV	R2, #0x100	;I3	Post-index
Loop	LDR	R4, [R2], #1	;I4	; (Step1) R4←[R2]. (Step2) R2←R2+1.
	ADD	R5, R4, R0	;I5	; R5←R4+R0 Post-index
	STR	R5, [R1], #1	;I6	; (Step1) [R1] ←R5. (Step2) R1←R1+1
	BNE	Loop	;I7	; Branch on Not Equal. (Z ≠ 1)
	SUB	R0, R0, #1	;I8	; R0 = R0 - 1
	NOP		;I9	
	STR	R1, [R1]	;I10	[R1] ←R1
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Figure Q1